Academic Helpdesk Agent

A multi-agent RAG assistant for IIT Jodhpur students

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1 What we built

We built a helpdesk agent that answers student questions about academic rules, curriculum, course slots and timetables, and writes (but never sends) an e-mail draft to a course instructor. It uses Google ADK as the agent framework, Qwen3 8B or Llama 3.1 8B (run through Ollama and LiteLLM) as the language model, `nomic-embed-text` for embeddings and ChromaDB as the vector store. If a question is not covered by our documents, the agent says it does not know and saves the question in a log for a human to answer. Every answer from the documents names its source.

We first built a single agent with all tools, and then a multi-agent version. We ran everything on a Google Colab T4 GPU.

2 System design

2.1 Documents

We collected **34 files**: 21 PDFs (academic regulations, the attendance and examination circulars, the academic calendar, programme list and the B.Tech./B.S./M.Sc. curriculum booklets) and 13 CSV sheets (department course lists and the slot system). The PDFs and sheets were split into **3,171 chunks** and stored in ChromaDB with their file name and page number, which is used for citations. A script (`prepare_csvs.py`) also merges the department sheets into three tables: `timetable.csv`, `courses.csv` and `faculty.csv`.

2.2 Tools

Tool	What it does
`search_documents`	Finds the most relevant chunks in ChromaDB and returns them with file name and page.
`lookup_timetable`	Gives the slot and lecture timing of a course from `timetable.csv`.
`list_courses`	Counts or lists courses by department, level and type from `courses.csv`.
`get_instructor_contact`	Gives the instructor's name and e-mail from `faculty.csv`.
`draft_email`	Writes an e-mail draft and saves it in `drafts/`. It is never sent.
`log_unanswered`	Saves a question the agent cannot answer in `logs/unanswered_questions.csv`. The helpdesk also logs every refusal automatically.

Table 1: The six tools.

2.3 Agents

In the multi-agent version, a *helpdesk orchestrator* reads the question and hands it to one of three agents (Figure 1). The single-agent version gives all six tools to one agent.

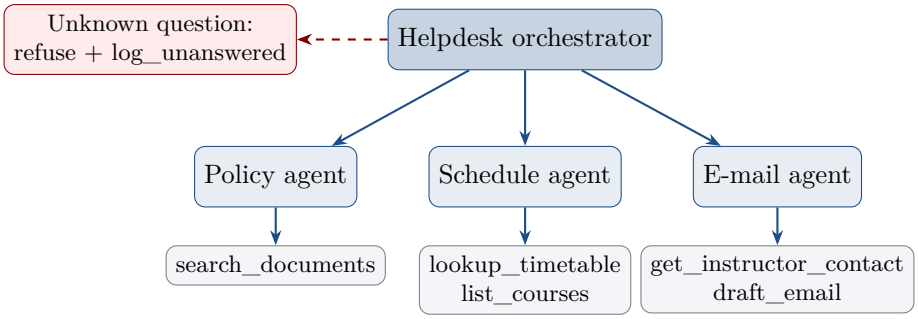


Figure 1: Multi-agent design.

2.4 Refusing and logging

For questions outside our documents the agent must reply: *"I'm sorry, I don't know the answer to that. I have logged your question so the academic office can help you."* The helpdesk code logs the question automatically even if the model forgets to call `log_unanswered`.

3 How we tested

We wrote **32 questions** (file `questions.csv`). For each one the file stores the expected agent, the expected source file and keywords the correct answer must contain.

Type	Example	Count
Policy (P)	What is the attendance requirement to sit in the Major exam?	13
Schedule (S)	What is the slot for the course Autonomous Systems?	7
E-mail (E)	Write an email to the instructor of CSL7650 that I need leave for 2 days.	4
Not in our data (N)	Slot of "Quantum Basket Weaving"; e-mail to instructor of CSL7570	2
Out of scope (O)	What are the available dining options on the campus?	6

Table 2: The 32 test questions.

Every question was run **3 times** in three set-ups: (1) multi-agent with Qwen3 8B, (2) multi-agent with Llama 3.1 8B (the model swap) and (3) single-agent with Qwen3 8B. Each question used a fresh session. The measures are:

- **Accuracy:** the answer contains the expected facts. Accuracy, citation and groundedness are averaged over the 24 questions that can be answered from our data (P, S and E types).
- **Citation present:** the answer names a source.
- **Groundedness:** the cited source is the expected file.
- **Correct refusal:** N and O questions get the "I don't know" reply and are logged.
- **False refusal:** an answerable question is refused.
- **Routing accuracy:** the orchestrator picks the expected agent.
- **Changes across runs:** questions whose answer text or correctness differs between the 3 runs.

4 Results

	Multi, Qwen3	**Multi, Llama 3.1**	**Single, Qwen3**
Accuracy (%)	63.9	50.0	75.0
Citation present (%)	80.6	62.5	87.5
Groundedness (%)	61.1	59.7	79.2
Out-of-scope questions refused (%)	100	100	100
Not-in-data questions refused (%)	50.0	100	100
False refusals (%)	19.4	33.3	12.5
Refusals logged (%)	100	100	100
Routing accuracy (%)	96.2	92.3	—
Questions with different answers in 3 runs	7	13	13
Questions whose correctness changed	1	6	0
Average time per question (s)	3.7	4.1	3.8

Table 3: Main results (32 questions × 3 runs).

Accuracy (%)	**Multi, Qwen3**	**Multi, Llama 3.1**	**Single, Qwen3**
Policy (13 questions)	53.8	59.0	53.8
Schedule (7)	76.2	47.6	100.0
E-mail (4)	75.0	25.0	100.0

Table 4: Accuracy by question type.

5 What we found

1. **Out-of-scope questions are handled well.** All six out-of-scope questions were refused in every run of every set-up, and every refusal was logged.

2. **The single agent did best; multi-agent did not help.** Single-agent Qwen3 reached 75.0% accuracy against 63.9% for multi-agent Qwen3, with fewer false refusals (12.5% vs. 19.4%) and similar speed. It answered every schedule and e-mail question correctly (Table **??**). The extra routing step in the multi-agent version adds one more place to go wrong, for example P04 was never passed to the policy agent.

3. **Qwen3 often skips the search tools.** For the 72 answers to answerable questions in multi-agent Qwen3, 25 involved no search or lookup at all: 14 were wrong refusals and 11 were answers given from memory. The single agent did this 12 times and Llama 3.1 11 times (all refusals). In multi-agent Qwen3, all 14 wrong refusals happened *before* any search was tried (P04, P10, P12, S07 and E04).

4. **Answers from memory can look correct but are made up.** In multi-agent Qwen3, question P05 (missed Major exam because of illness) was answered without a search. It cited `academic_regulations.pdf`, which is not one of our files, and said to "provide medical proof". The single agent did search and found the real rule (apply to the Dean

(Academics) through the instructor within ten days). For S03 the model invented "slot 3, Mondays and Wednesdays, room 205" in two of three runs, citing "timetable.csv, page 15". For S06 it said the slot was "not available", but the slot exists (F). The saved check still counted P05 and S03 as correct, which is why we added the checked groundedness row.

5. **Qwen3 is better overall, but Llama 3.1 reads documents well.** Qwen3 is ahead because of the tool-based schedule and e-mail questions (Table **??**). On policy questions Llama 3.1 is slightly better (59.0% vs. 53.8%): it answered P04 (6 semesters) and P12 (2 hours) in all runs, and P10 (course topics) in two runs, while both Qwen3 set-ups refused them. However, Llama 3.1 refused a third of the answerable questions (33.3%), sent P05 to the e-mail agent in every run, and was unstable (see Section 6).
6. **Some questions failed in every set-up.** P07, P08 and P09 were wrong everywhere. For P09 (start of the Major exam) nobody found the date in the academic calendar; the answers only say it is at the end of the semester. For P08 (last day to add or drop), the set-ups that searched answered "10 August 2026" from the calendar, which does not match the expected date in `questions.csv`; multi-agent Qwen3 answered "12 May 2027". For P07 (F grade) Qwen3 answered without searching and Llama 3.1 refused.
7. **Not-in-data questions.** Multi-agent Qwen3 refused CSL7570 correctly. For "Quantum Basket Weaving" it replied that the slot is "not available in the records" and cited `timetable.csv`, instead of giving the standard refusal. It did not invent a slot, but the answer is not logged as a refusal, so it counts as a failure. The other two set-ups refused both questions.

6 Questions whose answers changed across runs

Table 5 shows every question for which the answer text was not the same in all 3 runs. The number is how many different answers we saw (1 means identical every time). A star means the answer was correct in some runs and wrong in others.

- Most changes are only different wording, which is fine. E-mail drafts always differ because the model writes them freely.
- **Llama 3.1 changed correctness on six questions** (P02, P10, P13, S05, E01, E03). For example, for S05 it said "72 open electives" in one run and "5" in another, and refused in the third; for E03 it gave a wrong address (`ghari@iitj.ac.in`) once and the right one (`gharit@iitj.ac.in`) in the other runs.
- **Multi-agent Qwen3 changed correctness on one question** (S07): it answered once and refused twice. Its S03 answer also changed, with two runs inventing a timetable, but the saved check marked all three runs correct.
- **Single-agent Qwen3 never changed correctness**, although the wording changed on 13 questions.

7 Conclusion

The helpdesk answers questions from our 34 documents, cites its sources, drafts e-mails for review, and refuses and logs every out-of-scope question. In our tests the single agent with Qwen3 8B was the most accurate (75.0%) and the most stable, the multi-agent version was less accurate (63.9%), and Llama 3.1 8B was weaker (50.0%) and less stable. The main weakness of Qwen3 is that it sometimes refuses or answers without first searching, which leads to false refusals and a few answers with made-up sources.

ID	Question	Multi, Qwen3	Multi, Llama 3.1	Single, Qwen3
P01	Attendance to sit in Major exam	3	2	2
P02	Open elective credits (B.Tech.)	2	**2***	2
P03	Total credits, CSE B.Tech.	1	2	2
P05	Missed Major exam (sick)	2	1	3
P06	Maximum credits per semester	1	1	2
P07	Meaning of F grade, how to clear	1	1	2
P08	Last date to add/drop a course	1	2	2
P09	Start date of Major exam	1	3	3
P10	Topics in Autonomous Systems	1	**2***	1
P13	Absent more than four weeks	1	**2***	2
S03	CSL7650 lecture times	3	3	1
S05	Open electives offered	1	**3***	1
S07	Slot and timing, Operating Systems	**2***	2	1
E01	Leave e-mail, CSL7650	3	**3***	3
E02	Make-up exam e-mail, Autonomous Systems	2	1	3
E03	E-mail ID of CSL7650 instructor	1	**2***	2
E04	Sick-leave e-mail, Operating Systems	1	3	3

Table 5: Number of different answers in 3 runs (* = correctness also changed).

A Result for each question

Percentage of the 3 runs in which the answer was correct (re-scored). For N and O questions, "correct" means the question was refused.

ID	Question	Multi, Qwen3	Multi, Llama 3.1	Single, Qwen3
P01	Attendance to sit in Major exam	100	100	100
P02	Open elective credits (B.Tech.)	100	67	100
P03	Total credits, CSE B.Tech.	100	100	100
P04	Minimum duration of residence	0	100	0
P05	Missed Major exam (sick)	100	0	100
P06	Maximum credits per semester	100	100	100
P07	Meaning of F grade, how to clear	0	0	0
P08	Last date to add/drop a course	0	0	0
P09	Start date of Major exam	0	0	0
P10	Topics in Autonomous Systems	0	67	0
P11	Minimum weightage of Major exam	100	100	100
P12	Length of Minor exam	0	100	0
P13	Absent more than four weeks	100	33	100
S01	Slot of Autonomous Systems	100	100	100
S02	400-level Humanities electives	100	0	100
S03	CSL7650 lecture times	100	100	100
S04	Computer Science electives floated	100	0	100
S05	Open electives offered	100	33	100

ID	Question	Multi, Qwen3	Multi, Llama 3.1	Single, Qwen3
S06	Slot of Intro to Cyber Physical Systems	0	100	100
S07	Slot and timing, Operating Systems	33	0	100
E01	Leave e-mail, CSL7650	100	33	100
E02	Make-up exam e-mail, Autonomous Systems	100	0	100
E03	E-mail ID of CSL7650 instructor	100	67	100
E04	Sick-leave e-mail, Operating Systems	0	0	100
N01	Leave e-mail, CSL7570 (no such course)	100	100	100
N02	Slot of “Quantum Basket Weaving”	0	100	100
O01	Dining options on campus	100	100	100
O02	Reaching campus from city centre	100	100	100
O03	Hostel mess menu	100	100	100
O04	Cricket world cup winner	100	100	100
O05	Restaurant near campus	100	100	100
O06	Weather in Jodhpur	100	100	100